

American International University-Bangladesh (AIUB)
Department of Computer Science
Faculty of Science & Technology (FST)

E-WELFARE SYSTEM

A Software Engineering Project Submitted
By

Semester: Spring 24-25		Section: P	Group Number:01	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
01	ARPON PAUL AMIT	23-50158-1	20%	
02	MD TANJIM HOSSAIN	23-51464-1	20%	
03	PRITHA BISWAS PRAMA	22-46161-1	20%	
04	MD. SHAHRIAR CHY SHAJID	22-46645-1	20%	
05	FARHINA PARVIN SWARNA	22-47307-1	20%	

The project will be evaluated for the following Course Outcomes

CO3: Select appropriate software engineering models, project management roles, and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation Regarding Process Model Selection	[5Marks]
	Analysis of the impact of societal, health, safety, legal, and cultural issues	[5Marks]
CO4: Develop a project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]
	Total Marks	
	Develop the project plan, its components of the proposed software products	[5Marks]
	Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]
CO5: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in the computer science and engineering domain	Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]
	Total Marks	
Taking project responsibility: perform assigned tasks on time independently		[5 Marks]

Contribution to project group meetings, sharing fruitful ideas	[5Marks]	
Positive attitude towards group work, collaboration, compromise, helping others to understand their project work responsibility	[5Marks]	
Showing respect and value towards other team member's opinion	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: ARPON PAUL AMIT Student ID: 23-50158-1 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 _____ Arpon _____ Signature of the Student
Student Name: MD TANJIM HOSSAIN Student ID: 23-51464-1 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 _____ Tanjim _____ Signature of the Student
Student Name: PRITHA BISWAS PRAMA Student ID: 22-46161-1 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 _____ Prama _____ Signature of the Student
Student Name: MD. SHAHRIAR CHY SHAJID Student ID: 22-46645-1 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2

Signature of the Student

Student Name: FARHINA PARVIN SWARNA
Student ID: 22-47307-1

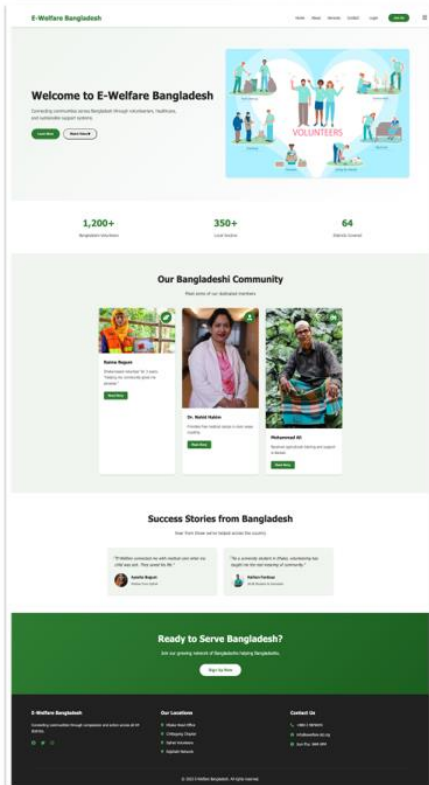
Contribution in Percentage (20%):

Contribution in the Project:

- Contribution Description 1
- Contribution Description 2

Signature of the Student

1. System Interface Design (UI/UX):



E-Welfare Responsive Design

- Wider content
- Stacked Cards

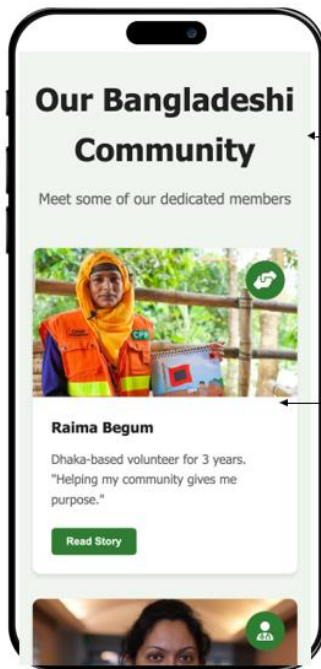
Desktop Version- Full Navigation

(1973x1080 standard for all desktop size screen)



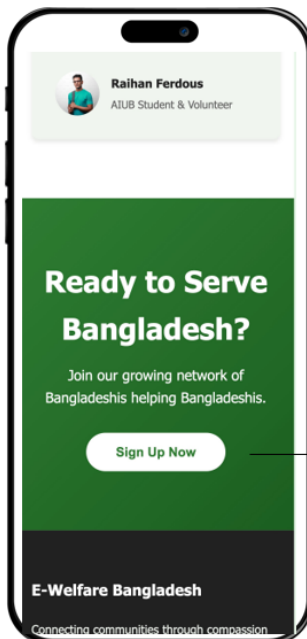
Stable Hamburger Icon

User Scrolling Content



Smooth Navigated UI

- Stacked Cards
- Dynamic Window to fit any screen

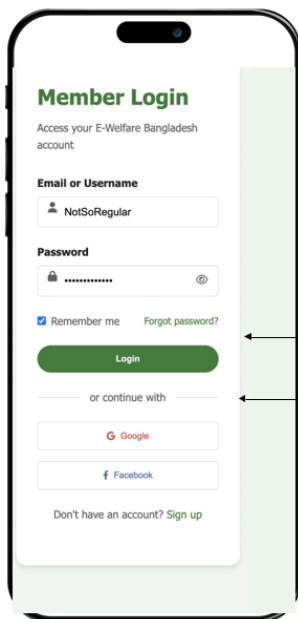


One [click](#) to Registration

 A mobile app screen showing the "Join E-Welfare Bangladesh" registration form. The form includes fields for Full Name, Username, Email, and Phone Number. It also has dropdown menus for "I want to join as:" (with a "Select role" option) and "District" (with a "Select District" option). A "Register Now" button is at the bottom. A close button (X) is in the top right corner.

- Blank fields trigger error: "Please fill in this field"
- Email format validated (@ and domain check)
- Phone number requires 11 digits (Bangladeshi format)
- Password strength indicator

- User can close the modal (click [X] or outside)
- Hamburger menu remains accessible
- All navigation links functional (Home/About/Services)



Member Login

Access your E-Welfare Bangladesh account.

Email or Username

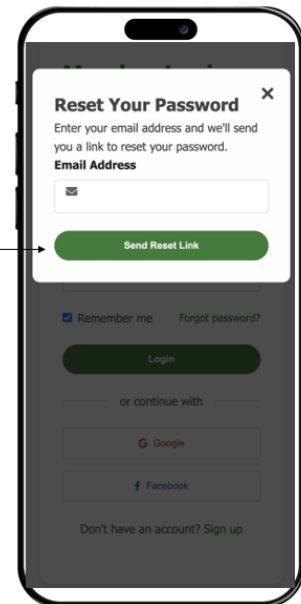
Password

☒ Remember me [Forgot password?](#)

or continue with

Don't have an account? [Sign up](#)

- Email/Phone + Password fields
- "Remember Me" checkbox (30-day session)
- Error handling:
 - ❌ "Invalid credentials" (generic for security)
 - 🔄 "Resend verification link" (if email unverified)
- Click "Forgot Password?" → Modal opens
- Input registered email → "Send Code" button
- Social Login (Google/Facebook)
- OAuth buttons with platform icons
- Permissions:
 - Basic profile (name/email)



Reset Your Password ✕

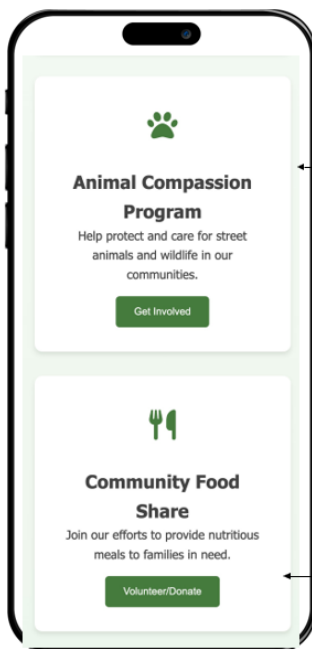
Enter your email address and we'll send you a link to reset your password.

Email Address

☒ Remember me [Forgot password?](#)

or continue with

Don't have an account? [Sign up](#)





Animal Compassion Program

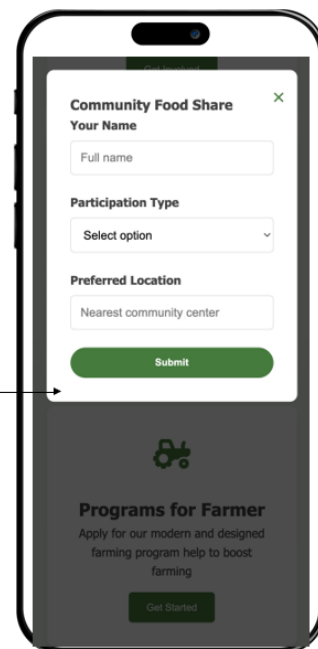
Help protect and care for street animals and wildlife in our communities.



Community Food Share

Join our efforts to provide nutritious meals to families in need.

- Service Discovery
- Click Action Flow
- Logged-in Users:
 - Click → Opens service-specific form (modal/popup)
 - Pre-filled with profile data (name/contact)
- Service-Specific Forms
- Email/SMS confirmation



Community Food Share ✕

Your Name

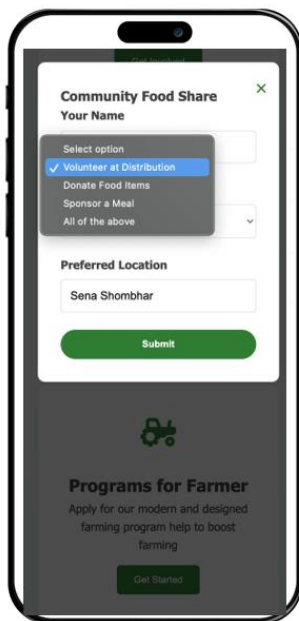
Participation Type

Preferred Location



Programs for Farmer

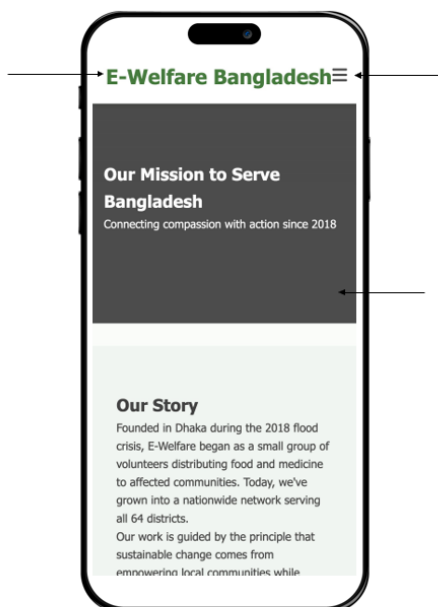
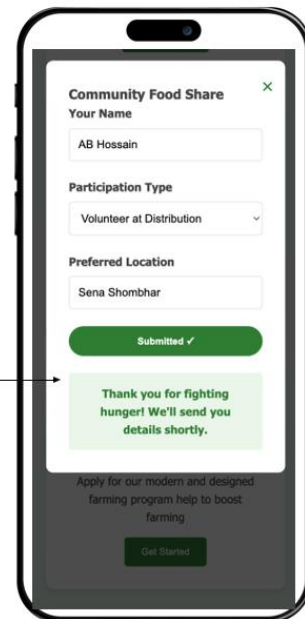
Apply for our modern and designed farming program help to boost farming



✓ Icon: Large green checkmark

🍌 Color: White text on (brand green) with subtle shadow

📱 Responsive: Adapts to modals/full-page (desktop & mobile)



- ✓ **Success:** Green header_with checkmark icon
- ⚠️ **Pending/Info:** Blue header_with info icon
- 📱 **Same layout** for modals, full-page messages, and mobile
- Smooth Transitions
- Confusion Checks
- No Dead-Ends (Hamburg Keeps on the top left)

2. Test Cases & Test Plan:

1.Module: Login Session:

The system shall allow secure login with valid credentials, block accounts after three failed attempts, and log out inactive users after 30 minutes.

Test Case ID: LS BB 01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: LS_BB_01 Test Priority (Low, Medium, High):Medium Module Name: Login Session			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify login with valid username and password				
Description: Test website login page				
Precondition: User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1.Go to the website		Login page loads successfully	As expected	Pass
2.Enter username	Username: Arpon	Username field accepts input	As expected	Pass
3.Enter password	Password: Arpon123@	Password field accepts input	As expected	Pass
4.Click submit		User is redirected to dashboard, session is active	As expected	Pass
Post Condition: User is validated with database and successfully logged into account. The account session details are logged in the database.				

Test Case ID: LS_BB_02

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: LS_BB_02 Test Priority (Low, Medium, High): Medium Module Name: Login Session			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify login with invalid password				
Description: Test login failure with incorrect password				
Precondition: User must have a registered username				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1.Go to the website		Login page loads successfully	As expected	Pass
2.Enter username	Username: Arpon	Username field accepts input	As expected	Pass
3.Enter incorrect password	Password: Arpon123	Password field accepts input	As expected	Pass
4.Click submit		Error message: "Invalid password"; account blocked after 3 attempts	As expected	Pass
Post Condition: User remains logged out, account temporarily blocked after 3 failed attempts.				

Test Case ID: LS_BB_03

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: LS_BB_03 Test Priority (Low, Medium, High): Medium Module Name: Login Session			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify login timeout with inactive session				
Description: Test session timeout when user remains inactive for too long				
Precondition: User must be logged into the system				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1.Log in with valid credentials	Username: Arpon, Password: Arpon123@	User logs in successfully	As expected	Pass
2.Remain inactive for 30 minutes		Session expires; user is logged out	As expected	Pass
3.Attempt to access dashboard	Password: Arpon123	Redirected to login page with message:"Session expired"	As expected	Pass
Post Condition: User remains logged out, account temporarily blocked after 3 failed attempts.				

2. Module: Sign Up

The system shall allow users to sign up with valid details and activate accounts via email verification.

Test Case ID: SU_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: SU_BB_01 Test Priority (Low, Medium, High): High Module Name: Sign Up			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify successful user sign-up				
Description: Test the sign-up process with valid inputs				
Precondition: User must not have an existing account				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to sign-up page		Sign-up page loads successfully	As expected	Pass
2. Enter full name	Name: Arpon	Name field accepts input	As expected	Pass
3. Enter email and password	Email:arponamit.55@gmail.com, Password: Arpon123@	Fields accept valid input	As expected	Pass
4. Enter contact and select role	Contact: 8801712345678, Role: Donor	Fields accept valid input	As expected	Pass
5. Click submit		Verification email sent; account activated after verification	As expected	Pass
Post Condition: User account is created and verified.				

3. Module: Sign In

The system shall allow secure sign-in with email, password, and MFA OTP for verified accounts.

Test Case ID: SI_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: SI_BB_01 Test Priority (Low, Medium, High): High Module Name: Sign In			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify sign-in with MFA				
Description: Test sign-in process with multi-factor authentication				
Precondition: User must have a verified account				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to sign-in page		Sign-in page loads successfully	As expected	Pass
2. Enter email and password	Email: arponamit.55@gmail.com, Password: Arpon123@	Credentials accepted	As expected	Pass
3.Enable MFA		MFA prompt displayed	As expected	Pass
4. Enter OTP	OTP: 45238	OTP validated; user redirected to dashboard	As expected	Pass
5. Click submit		User logs in with active session	As expected	Pass
Post Condition: User is logged in with active session.				

4. Module: Forgot Password:

The system shall allow password reset via email with a link valid for 15 minutes.

Test Case ID: FP_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: FP_BB_01 Test Priority (Low, Medium, High): Medium Module Name: Forgot Password			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify password reset process				
Description: Test the forgot password functionality				
Precondition: User must have a registered account				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to forgot password page		Forgot password page loads successfully	As expected	Pass
2. Enter registered email	E-mail: arponamit.55@gmail.com	Email field accepts input	As expected	Pass
3. Click submit		Reset link sent to email, valid for 15 minutes	As expected	Pass
Post Condition: User receives a reset link and can update password.				

5. Module: Profile Check and Update

The system shall allow logged-in users to update profile details with password verification.

Test Case ID: PCU_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: PCU_BB_01 Test Priority (Low, Medium, High): Medium Module Name: Profile Check and Update			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify profile update				
Description: Test updating user profile details				
Precondition: User must be logged in				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1.Go to profile page		Profile page loads successfully	As expected	Pass
2.Update email	New Email: arponamit.55@gmail.com	Email field accepts input	As expected	Pass
3.Enter current password	Current Password: Arpon@123	Password validated	As expected	Pass
4.Click save		Email updated after verification; confirmation email sent	As expected	Pass
Post Condition: Profile updated and verified.				

6. Module: Donation Submission

The system shall allow donors to submit cash donations via a linked payment method.

Test Case ID: DS_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: DS_BB_01 Test Priority (Low, Medium, High): High Module Name: Donation Submission			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify cash donation submission				
Description: Test submitting a cash donation				
Precondition: User must be logged in as a donor				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1.Go to donation page		Donation page loads successfully	As expected	Pass
2.Select cash donation		Cash donation form displayed	As expected	Pass
3.Enter amount and payment	Amount: 1000, Payment: Card	Fields accept valid input	As expected	Pass
4.Click perform		Payment confirmed; donation recorded	As expected	Pass
Post Condition: Donation is successfully submitted.				

7. Module: Donation Tracking

The system shall allow users to track donation status and generate receipts using donation ID.

Test Case ID: DT_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: DT_BB_01 Test Priority (Low, Medium, High): Medium Module Name: Donation Tracking			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify donation tracking				
Description: Test tracking a submitted donation				
Precondition: User must have made a donation				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to donation history		Donation history page loads successfully	As expected	Pass
2. Select donation	Donation ID: D123	Donation details displayed	As expected	Pass
3. Click track		Status displayed (e.g., in transit); receipt generated	As expected	Pass
Post Condition: Donation status and receipt available.				

8. Module: Donation Distribution

The system shall allow assignment of donations to beneficiaries with status updates.

Test Case ID: DD_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: DD_BB_01 Test Priority (Low, Medium, High): High Module Name: Donation Distribution			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify donation distribution				
Description: Test assigning donations to beneficiaries				
Precondition: Donations must be available				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to distribution page		Distribution page loads successfully	As expected	Pass
2. Select donation and beneficiary	Donation: D123, Beneficiary: B001	Selection accepted	As expected	Pass
3Click assign		Donation assigned; status updated	As expected	Pass
Post Condition: Donation distributed to beneficiary.				

9. Module: Free Medical Services

The system shall allow verified underprivileged users to request free medical services with resource allocation.

Test Case ID: FMS_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: FMS_BB_01 Test Priority (Low, Medium, High): High Module Name: Free Medical Services			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify free medical service request				
Description: Test requesting free medical services				
Precondition: User must be verified as underprivileged				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1.Go to medical services page		Medical services page Grown	As expected	Pass
2.Enter condition	Condition: Fever	Condition field accepts input	As expected	Pass
3.Select delivery method	Delivery: Home	Delivery option selected	As expected	Pass
4.Click submit		Request accepted; resources allocated	As expected	Pass
Post Condition: Request processed and resources allocated.				

10. Module: Paid Medical Services

The system shall allow users with linked payment methods to book and pay for medical consultations.

Test Case ID: PMS_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: PMS_BB_01 Test Priority (Low, Medium, High): Medium Module Name: Paid Medical Services			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify paid medical service booking				
Description: Test booking a paid consultation				
Precondition: User must have a linked payment method				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to paid services page		Paid services page loads successfully	As expected	Pass
2. Select service	Service: Consultation	Service selected	As expected	Pass
3. Enter payment details	Payment: Card	Payment details accepted	As expected	Pass
4. Click book		Booking confirmed; payment processed	As expected	Pass
Post Condition: Appointment booked and payment processed.				

11. Module: Emergency Contact System

The system shall allow emergency requests with location and crisis details, routing to volunteers.

Test Case ID: ECS_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: ECS_BB_01 Test Priority (Low, Medium, High): High Module Name: Emergency Contact System			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify emergency contact request				
Description: Test submitting an emergency request				
Precondition: System must be operational				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Access helpline		Helpline page loads successfully	As expected	Pass
2. Enter location and crisis	Location: Feni, Crisis: Flood	Fields accept input	As expected	Pass
3.Click submit		Request routed to volunteers; confirmation received	As expected	Pass
Post Condition: Help dispatched to location.				

12. Module: Crisis Resource Allocation

The system shall allow allocation of resources to registered crisis areas with status updates.

Test Case ID: CRA_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: CRA_BB_01 Test Priority (Low, Medium, High): High Module Name: Crisis Resource Allocation			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify crisis resource allocation				
Description: Test allocating resources to a crisis area Precondition: Crisis event must be registered				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to allocation page		Allocation page loads successfully	As expected	Pass
2. Select resource and quantity	Location: Feni, Crisis: Flood	Resource: Food, Quantity: 100, Location: Feni	As expected	Pass
3. Click allocate		Resources allocated; status updated	As expected	Pass
Post Condition: Resources delivered to crisis area.				

13. Module: Volunteer Recruitment

The system shall allow new volunteers to register with skills and availability details.

Test Case ID: VR_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: VR_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Volunteer Recruitment			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify volunteer registration				
Description: Test registering a new volunteer Precondition: User must not be a registered volunteer				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to volunteer page		Volunteer page loads successfully	As expected	Pass
2. Enter skills and availability	Skills: Teaching, Availability: Weekends	Fields accept input	As expected	Pass
3. Click register		Registration confirmed; volunteer profile created	As expected	Pass
Post Condition: Volunteer profile created.				

14. Module: Volunteer Task Management

The system shall allow task assignment to volunteers and progress updates.

Test Case ID: VTM_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: VTM_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Volunteer Task Management			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify task assignment to volunteer				
Description: Test assigning and updating a task Precondition: Volunteer must be registered				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to task page		Task page loads successfully	As expected	Pass
2. Assign task to volunteer	Task: Food Distribution	Task assigned to volunteer	As expected	Pass
3. Update progress	Progress: 50%	Progress updated in system	As expected	Pass
Post Condition: Task status updated.				

15. Module: Job Opportunities

The system shall allow users to apply for verified employer job listings.

Test Case ID: JO_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: JO_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Job Opportunities			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify job application process				
Description: Test applying for a job listing Precondition: Employer must be verified				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to job page		Job page loads successfully	As expected	Pass
2. Apply with profile	Job: NGO Worker	Application form populated with profile data	As expected	Pass
3. Click submit		Application submitted and recorded	As expected	Pass
Post Condition: Application recorded.				

16. Module: Skill Development

The system shall allow users to enroll in available skill development courses.

Test Case ID: SD_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: SD_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Skill Development			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify enrollment in training module				
Description: Test enrolling in a skill development course Precondition: Training modules must be available				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to training page		Training page loads successfully	As expected	Pass
2. Select module	Module: Farming, Mode: Online	Selection accepted	As expected	Pass
3. Click enroll		Enrollment confirmed; user added to module	As expected	Pass
Post Condition: User enrolled in module.				

17. Module: Tree Plantation Programs

The system shall allow users to register for scheduled tree plantation events.

Test Case ID: TPP_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: TPP_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Tree Plantation Programs			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify tree plantation event registration				
Description: Test registering for a tree plantation event Precondition: Event must be scheduled				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to event page		Event page loads successfully	As expected	Pass
2. Enter location	Location: Dhaka	Location field accepts input	As expected	Pass
3. Click register		Registration confirmed; user added to event	As expected	Pass
Post Condition: User registered for event.				

18. Module: Community Cleaning

The system shall allow users to register for scheduled community cleaning events.

Test Case ID: CC_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: CC_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Community Cleaning			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify community cleaning event registration				
Description: Test registering for a cleaning event Precondition: Event must be scheduled				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to event page		Event page loads successfully	As expected	Pass
2. Enter availability	Availability: Weekend	Availability field accepts input	As expected	Pass
3. Click register		Registration confirmed; user added to event	As expected	Pass
Post Condition: User registered for event.				

19. Module: Surplus Resource Collection

The system shall allow reporting of surplus resources for collection by volunteers/NGOs.

Test Case ID: SRC_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: SRC_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Surplus Resource Collection			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify surplus resource reporting				
Description: Test reporting surplus resources Precondition: Volunteers/NGOs must be available				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to surplus page		Surplus page loads successfully	As expected	Pass
2. Enter resource details	Resource: Food, Quantity: 50, Pickup: Dhaka	Fields accept input	As expected	Pass
3. Click submit		Collection assigned to volunteer/NGO	As expected	Pass
Post Condition: Collection scheduled.				

20. Module: Surplus Resource Distribution

The system shall allow distribution of collected surplus resources to beneficiaries.

Test Case ID: SRD_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: SRD_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Surplus Resource Distribution			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify surplus resource distribution				
Description: Test distributing surplus resources Precondition: Resources must be collected				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to distribution page		Distribution page loads successfully	As expected	Pass
2. Select resource and beneficiary	Resource: Food, Quantity: 50, Location: Sylhet	Selection accepted	As expected	Pass
3. Click distribute		Resources distributed; status updated	As expected	Pass
Post Condition: Resources delivered to beneficiary.				

21. Module: Land Leasing

The system shall allow verified farmers to list and request land leases with defined terms.

Test Case ID: LL_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: LL_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Land Leasing			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify land leasing process				
Description: Test listing and requesting land lease Precondition: Farmer must be verified				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to leasing page		Leasing page loads successfully	As expected	Pass
2. List land	Size: 1 acre, Location: Jamalpur, Terms: 6 months	Fields accept input	As expected	Pass
3. Submit request		Lease agreement confirmed	As expected	Pass
Post Condition: Lease agreement finalized.				

22. Module: Direct Agricultural Trade

The system shall allow farmers to list and sell crops directly to customers with payment processing.

Test Case ID: DAT_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: DAT_BB_01 Test Priority (Low, Medium, High): High Module Name: Direct Agricultural Trade			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify crop sale transaction				
Description: Test selling crops directly				
Precondition: Farmer must have listed crops				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to trade page		Trade page loads successfully	As expected	Pass
2. List crop and price	Crop: Rice, Quantity: 100, Price: 5000	Fields accept input	As expected	Pass
3. Customer orders	Payment: Card	Order received	As expected	Pass
4. Click confirm		Sale completed; payment processed	As expected	Pass
Post Condition: Transaction completed.				

23. Module: Crop Management

The system shall allow farmers to input crop growth data and request analytics.

Test Case ID: CM_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: CM_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Crop Management			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify crop data input				
Description: Test entering crop growth data Precondition: Farmer must have active crops				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1 Go to crop page		Crop page loads successfully	As expected	Pass
2. Enter growth data	Planting Date: 2025-04-01, Harvest: 2025-08-01	Data recorded in system	As expected	Pass
3. Request analytics		Analytics generated and displayed	As expected	Pass
Post Condition: Data and analytics available.				

24. Module: Sponsored Advertisements

The system shall allow verified businesses to submit and display paid advertisements.

Test Case ID: SA_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: SA_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Sponsored Advertisements			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify ad submission				
Description: Test submitting an advertisement				
Precondition: Business must be verified				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to ad page		Ad page loads successfully	As expected	Pass
2. Enter ad details	Content: Banner, Duration: 1 month, Payment: 500	Fields accept input	As expected	Pass
3. Click submit		Ad approved and displayed on platform	As expected	Pass
Post Condition: Ad displayed on platform.				

25. Module: Revenue Tracking

The system shall allow tracking and allocation of revenue to initiatives.

Test Case ID: RT_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: RT_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Revenue Tracking			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify revenue allocation				
Description: Test tracking and allocating revenue				
Precondition: Revenue transactions must exist				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to revenue page		Revenue page loads successfully	As expected	Pass
2. View revenue	Revenue: 1000, Initiative: Medical	Revenue details displayed	As expected	Pass
3. Click allocate		Funds allocated to initiative	As expected	Pass
Post Condition: Funds allocated to initiative.				

26. Module: Property Listing

The system shall allow verified users to list properties for rent with defined terms.

Test Case ID: PL_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: PL_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Property Listing			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify property listing				
Description: Test listing a property for rent				
Precondition: User must own the property				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1 Go to property page		Property page loads successfully	As expected	Pass
2. Enter details	Price: 10000, Location: Dhaka, Terms: 1 month	Fields accept input	As expected	Pass
3. Click list		Property listing confirmed	As expected	Pass
Post Condition: Property listed.				

27. Module: Rental Transactions

The system shall allow users to book listed rental properties with payment processing.

Test Case ID: RT_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: RT_BB_01 Test Priority (Low, Medium, High): Medium Module Name: Rental Transactions			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify rental booking				
Description: Test booking a rental property				
Precondition: Property must be listed				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to rental page		Rental page loads successfully	As expected	Pass
2. Select property	Property: P001	Property selected	As expected	Pass
3. Enter payment	Payment: Card	Payment details accepted	As expected	Pass
4. Click book		Booking confirmed; payment processed	As expected	Pass
Post Condition: Rental booked.				

28. Module: Property Management

The system shall allow users to manage rental bookings and track income.

Test Case ID: PM_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: PM_BB_01 Test Priority (Low, Medium, High): Medium Module Name : Property Management			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify property update				
Description: Test managing rental bookings				
Precondition: Property must be listed				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1 . Go to management page		Management page loads successfully	As expected	Pass
2. Update booking	Booking: B001	Booking details updated	As expected	Pass
3. Track income	Income: 10000	Income recorded	As expected	Pass
Post Condition: Booking and income updated.				

29. Module: NGO Integration

The system shall allow verified NGOs to register with valid details and API key.

Test Case ID: NI_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: NI_BB_01 Test Priority (Low, Medium, High): High Module Name : NGO Integration			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify NGO registration				
Description: Test registering an NGO				
Precondition: NGO must be verified				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1 . Go to NGO page		NGO page loads successfully	As expected	Pass
2. Enter details	Name: NGO1, API Key: XYZ	Fields accept input	As expected	Pass
3. Click register		Registration confirmed; NGO integrated	As expected	Pass
Post Condition: NGO integrated.				

30. Module: Bank Integration

The system shall allow users to link payment methods via bank API securely.

Test Case ID: BI_BB_01

Project Name: E-Welfare System			Test Designed by: Arpon	
Test Case ID: BI_BB_01 Test Priority (Low, Medium, High): High Module Name : Bank Integration			Test Designed date: 27-06-2025	
			Test Executed by: Arpon	
			Test Execution date: 15-05-2025	
Test Title: Verify bank payment link				
Description: Test linking a payment method				
Precondition: Bank API must be available				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1 . Go to payment page		Payment page loads successfully	As expected	Pass
2. Enter card details	Card: 1234-5678	Card details accepted	As expected	Pass
3. Click link		Payment method linked successfully	As expected	Pass
Post Condition: Payment method linked.				

3. Project Schedule and Effort Estimation

3.1 Work Breakdown Structure (WBS):

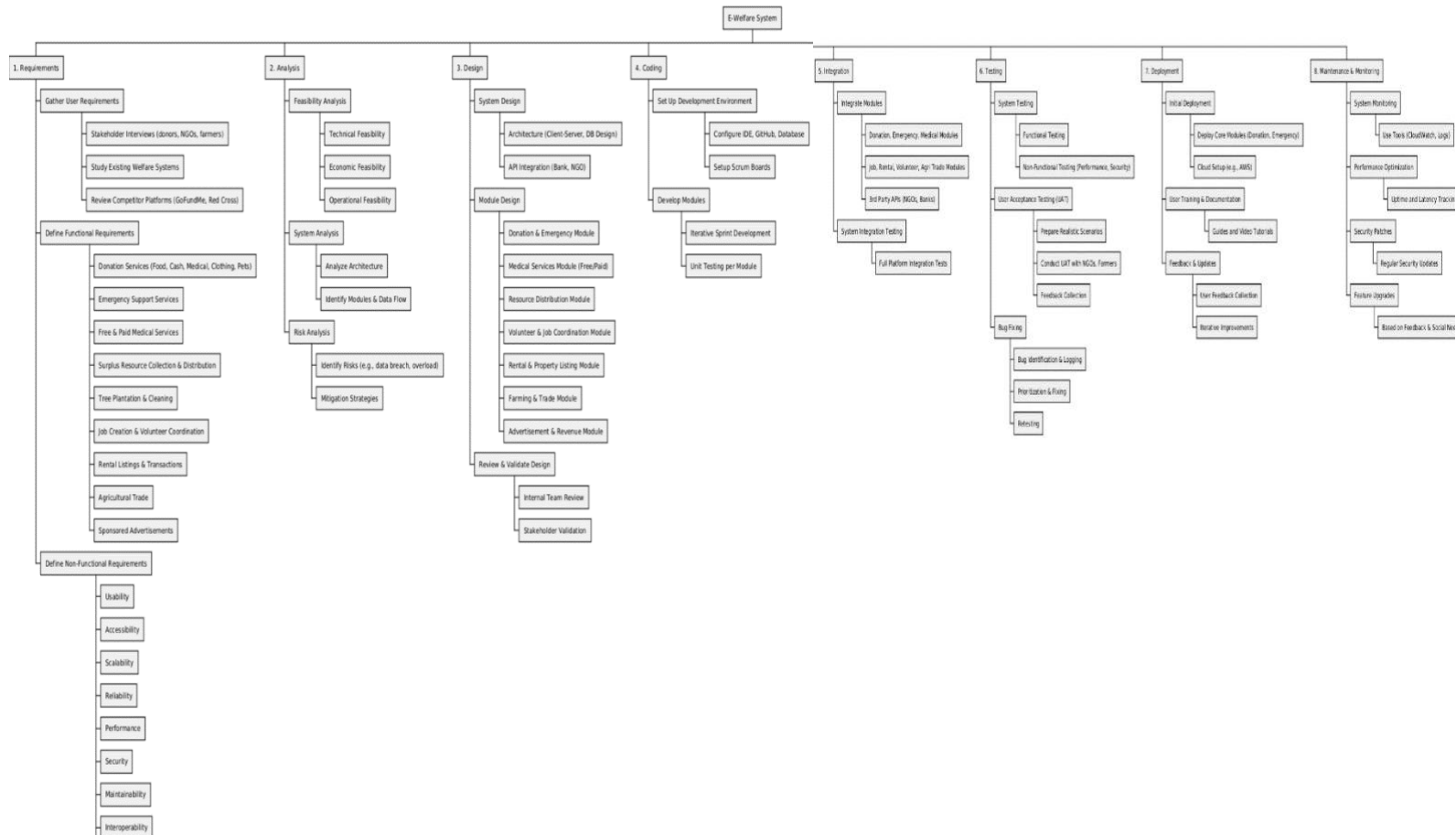


Fig 3.1 : Work Breakdown Structure (WBS) Diagram

3.2 COCOMO(Constructive Cost Model):

Software Project Type	Coefficient <Effort Factor>	P	T
Organic	2.4	1.05	0.38
Semi-detached	3.0	1.12	0.35
Embedded	3.6	1.20	0.32

Software Project Type=Organic

SLOC=16000

P=1.05

T=0.38

Coefficient<effort factor>=2.4

Effort=PM=Coefficient<effort factor> *(SLOC/1000)^P

=2.4*(22000/1000)^1.05

=61.624 $\approx$ 62 .00

Development time= DM = 2.50*(PM)^T

=2.50*(62.00)^0.38

=11.99 $\approx$ 12

Required number of people= $ST=PM/DM$
 $=62.00/12.00$
 $=5.16 \approx 5$

3.3 Gantt Chart:

E - Welfare System

Project start: **Mon, 3/10/2025**
Display week: **1**

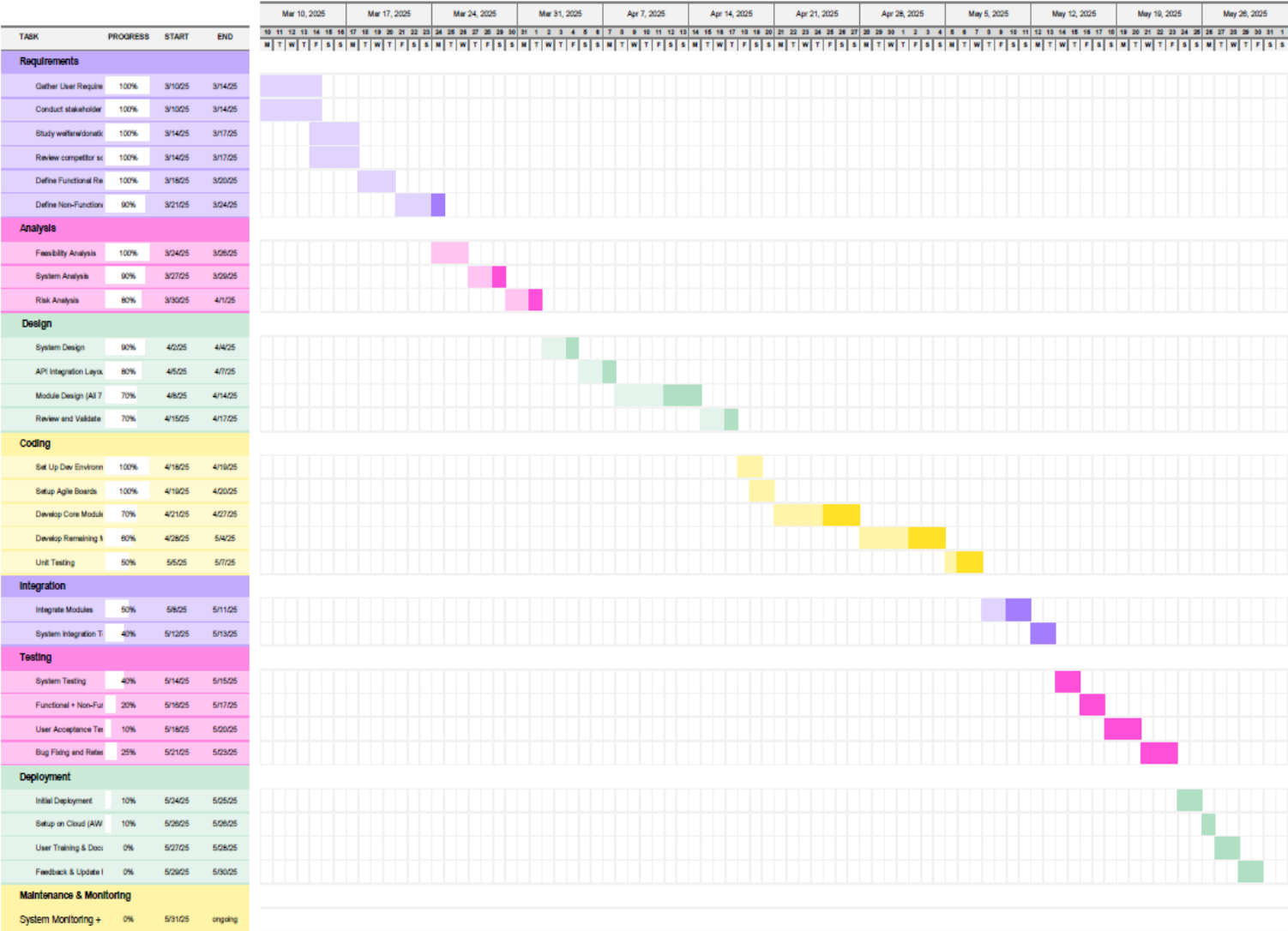


Fig 3.3 : Gantt Chart

4. Risk Management Plan:

ID	Description	Impact	Response	Level	Owner	Notes	Probability Level (%)
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R01	Server overload during high donation surges	Disrupts emergency aid delivery	Implement cloud scaling (e.g., AWS Auto Scaling)	Critical	System Integration Specialist	High risk during monsoon season (June-Aug)	80
R02	Data breaches compromising user privacy	Loss of trust and legal penalties	Use AES-256 encryption and conduct penetration testing	Critical	Security Specialist	Ongoing monitoring required as of 06/28/25	60
R03	Delayed deployment due to evolving requirements	Project timeline slippage	Conduct frequent sprint reviews	Medium	Scrum Master	Agile adjustments needed by next sprint	70
R04	Connectivity issues in rural areas	Limits farmer and beneficiary access	Develop lightweight, offline-capable interfaces	High	UI/UX Designer	Test in rural areas by 07/15/25	85
R05	Insufficient volunteer participation	Reduced crisis response capacity	Enhance outreach campaigns and incentives	High	Volunteer & Donor Coordinator	Critical during peak disaster periods	65
R06	Failed payment integrations with banks	Delays in donation processing	Test bank APIs thoroughly and retry mechanisms	High	System Integration Specialist	Review integration status by 07/01/25	60
R07	Inaccurate crop data affecting trade	Financial losses for farmers	Validate inputs and integrate market analytics	Medium	Development Team	Pilot testing ongoing as of 06/28/25	55
R08	Non-compliant ad content from businesses	Legal and reputational damage	Review submissions and enforce guidelines	Medium	Marketing & Outreach Coordinator	Initial review due by 07/10/25	40
R09	System downtime during maintenance	Interrupts critical services	Schedule maintenance during low-traffic periods	Critical	QA Team	Plan maintenance for off-peak hours	30

R10	Legal issues with property rental agreements	Financial and legal disputes	Verify ownership and ensure legal compliance	High	Risk Manager	Legal review needed by 07/15/25	35
R11	Low adoption by target users	Reduced system effectiveness	Simplify UI/UX and conduct rural campaigns	Medium	Marketing & Outreach Coordinator	Launch campaign by 07/01/25	60
R12	Resource shortages during crisis	Fails to meet beneficiary needs	Maintain buffer stocks and backup suppliers	Critical	Crisis Resource Allocation Team	Assess stocks by 06/30/25	75
R13	Inconsistent NGO data integration	Delays in coordination	Standardize API protocols and test regularly	Medium	System Integration Specialist	Test with NGOs by 07/05/25	50
R14	Delays in emergency response	Life-threatening delays in aid delivery	Prioritize routing algorithms and test loads	Critical	Development Team	Test under simulation by 07/10/25	80
R15	High energy consumption	Increased costs and environmental impact	Optimize with green hosting providers	Low	System Integration Specialist	Review energy usage by 08/01/25	40
R16	Misallocation of donations	Undermines donor trust	Implement eligibility checks and audits	Medium	Donation Distribution Team	Audit process due by 07/15/25	55
R17	Security vulnerabilities in MFA	Unauthorized access to accounts	Conduct security audits and update protocols	High	Security Specialist	Audit scheduled for 07/01/25	60
R18	Lack of scalability for new programs	Limits future growth	Design modular architecture and plan expansion	Medium	Development Team	Plan review by 07/20/25	50
R19	Poor performance under peak loads	Slows transaction processing	Optimize database queries and cache requests	High	QA Team	Performance test due by 07/10/25	55

R20	Resistance from farmers to adopt trade	Reduced agricultural engagement	Provide training and support for farmers	Low	Volunteer & Donor Coordinator	Training session planned for 07/15/25	60
R21	Inadequate training for volunteers	Inefficient task execution	Develop training modules and conduct sessions	Medium	Volunteer & Donor Coordinator	Training due by 07/10/25	50
R22	Delays in medical resource allocation	Health risks to beneficiaries	Streamline verification and prioritize requests	High	Development Team	Review process by 07/05/25	70
R23	Legal disputes over land leasing	Delays in farming initiatives	Conduct legal reviews and mediation	High	Risk Manager	Legal assessment by 07/15/25	45
R24	Insufficient funding from advertisements	Threatens financial sustainability	Diversify revenue sources and secure sponsors	High	Marketing & Outreach Coordinator	Proposal due by 07/20/25	55
R25	Loss of user data during system upgrades	Loss of critical records	Implement backup and rollback procedures	Critical	QA Team	Upgrade scheduled for 07/10/25, test backup	40

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing	Some errors in spelling and grammar. Some problems	Few errors in spelling and grammar. Presents most of the details in	Project report is complete and No errors in spelling and grammar. Consistently	

	organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	a logical flow of organization in definition, details, and example.	presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

Rubric for Project Assessment (CO4)

Marking Criteria	Marks Distribution (Maximum 3X5=15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					

CO5 [PO-i-2]: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in computer science and engineering domain.				
Assessment Attribute/Criteria	Missing/ Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Taking responsibility	Does not perform assigned tasks; often misses meetings and, when present, does not have anything constructive to say; relies on others to do the work;	Partially performs all assigned tasks; attends meetings irregularly and occasionally participates and hence not reliable;	Performs all assigned tasks; attends meetings regularly and usually participates effectively. generally reliable;	Performs all tasks very effectively; attends all meetings and participates enthusiastically; very reliable.
Contributions	Never provides useful ideas when participating in a group discussion	Rarely provides useful ideas when participating in a group discussion	Sometimes provides useful ideas when participating in a group discussion	Routinely provides useful ideas when participating in a group discussion
Collaboration and Ability to Compromise	Not cooperative, unable to compromise and disrupts the team process.	Sometimes cooperative, and rarely displays a positive attitude.	Usually cooperative, able to compromise and generally display positive attitude.	Always cooperative. Willingness to compromise. Always display positive attitude.
Valuing other team members (Working with others)	Often argues with teammates; doesn't let anyone else talk; occasional personal attacks and "put-downs"; wants to have things done his way and does not listen to alternate approaches.	Seldom listens to others' points of view; occasionally behaves in an oppressive manner; tries to force their own ideologies on other.	Generally, listens to others' points of view; always uses appropriate and respectful language; tries to make a definite effort to understand others' ideas.	Always listens to others and their ideas; helps them develop their ideas while giving them full credit; always helps the team reach a fair decision.